

Convex relaxation gaps and spatial certificates in nonlinear optimization

Abstract

Separate convexification of nonlinear terms can lose information about their shared variables. We develop a probability formulation of this loss and explain its relation to spatial optimality certificates. For bilinear functions with arbitrary coefficient signs, an elementary cut argument bounds the ratio of the McCormick gap to the scalar graph-hull gap by four times the square root of the largest induced edge density. A random-sign construction gives the matching order on each individual interaction graph. Sharper maximum-degree and bipartite bounds, exactness criteria, and the distinction between exact termwise convexification and recursive product relaxations identify the scope of these comparisons. The density order is also a consequence of earlier Schur-multiplier theory; the explicit transfer and proof are included to make that attribution precise.

1 Introduction

Convex relaxations support global optimization by providing bounds on nonlinear functions over a region. Their strength depends on how the model is divided into pieces. Even when each piece is replaced by its exact convex hull, the resulting bounds can be weaker than the hull of their sum: separate pieces may use incompatible representations of their common variables. Spatial subdivision introduces a second question, namely how many regions are needed before a specified bounding rule certifies a desired tolerance. These questions concern different mathematical objects. A pointwise envelope gap comparison is not an approximation ratio for a constrained optimization problem, and it does not by itself bound a branch-and-bound tree.

McCormick’s construction provides the basic recursive mechanism for factorable functions [4]. For multilinear functions, the convex hull of the scalar graph has a finite representation using box vertices. Luedtke, Namazifar, and Linderoth [3] compare this hull with termwise and recursive relaxations, and establish several exactness and positive-coefficient bounds. Boland et al. [1] show that signed bilinear gaps can grow with the square root of dimension and relate the comparison to cuts in induced interaction graphs.

We use the vertex representation to state the comparisons consistently. For positive multilinear functions on nonnegative boxes, a common probability distribution attains all upper term envelopes, so the entire gap question reduces to a single simultaneous rounding problem. For signed bilinear functions, an exact cut-range characterization instead permits graph density and degree bounds. The elementary density proof makes the constants explicit, while the accompanying Schur-multiplier transfer explains which asymptotic content was already implicit in established theory [2]. The framework also specifies the bounding oracle and the certified-region count needed for spatial lower bounds.

2 Envelope gaps as probability problems

2.1 The scalar graph and the specified termwise relaxation

Let $B = \prod_{i=1}^n [\ell_i, u_i]$ be a finite box. Fixed coordinates are removed when necessary. A function is *multiaffine* if it is affine in each coordinate separately; equivalently, it is a multilinear polynomial with no repeated variable in a monomial. For a continuous scalar function f on B , let $\text{vex}_B f$ be its greatest convex underestimator and $\text{cav}_B f$ its least concave overestimator. The vertical section of the convex hull of its graph is

$$\{y : (x, y) \in \text{conv}\{(z, f(z)) : z \in B\}\} = [\text{vex}_B f(x), \text{cav}_B f(x)].$$

Define the *scalar graph-hull gap* and, for a specified decomposition $f = \sum_{e \in E} f_e$, the *termwise gap* by

$$H_B(f; x) = \text{cav}_B f(x) - \text{vex}_B f(x), \quad (2.1)$$

$$T_B(f; x) = \sum_{e \in E} [\text{cav}_B f_e(x) - \text{vex}_B f_e(x)]. \quad (2.2)$$

The notation T_B always refers to the displayed decomposition. It is the width obtained by giving each scalar factor its exact graph hull and imposing $y = \sum_e y_e$. It can change when a factor is split into smaller pieces, whereas H_B does not. Always,

$$0 \leq H_B(f; x) \leq T_B(f; x), \quad (2.3)$$

because the sum of convex underestimators is a convex underestimator of the sum, and similarly for concave overestimators. Ratios are taken only where $H_B(f; x) > 0$; gap inequalities apply at every point.

For a positive multilinear polynomial, the factors are its displayed monomials $a_e \prod_{i \in e} x_i$, with $a_e \geq 0$. For a signed bilinear function, they are the signed products $a_{ij} x_i x_j$. A bilinear factor has the familiar four-inequality McCormick hull. For a higher-degree factor, its exact hull and a recursive McCormick relaxation are different objects. A recursive product relaxation is exact for one monomial on a unit box, but need not be exact on a general positive box [3, Sections 2–3]. A bound for (2.2) on that box therefore does not automatically bound the gap of a recursive bilinear factorization.

Adding an affine function to a factor, or adding a separate affine factor, changes neither gap. An affine bijection between boxes preserves both envelopes after composition and hence preserves the gaps for the transformed factors. These facts follow from the graph-hull representation.

2.2 Vertex distributions and compatible marginals

Let $\mathcal{V}(B)$ denote the vertices of B , and let $\mathcal{P}_B(x)$ be the distributions on $\mathcal{V}(B)$ with mean x . On the unit cube these are joint Bernoulli distributions with the prescribed singleton means; coordinates need not be independent.

Lemma 2.1 (Vertex representation). *For a multiaffine f on B ,*

$$\text{vex}_B f(x) = \min_{P \in \mathcal{P}_B(x)} \mathbb{E}_P f(X), \quad (2.4)$$

$$\text{cav}_B f(x) = \max_{P \in \mathcal{P}_B(x)} \mathbb{E}_P f(X). \quad (2.5)$$

Both extrema are attained. The envelopes are continuous piecewise-affine functions on B .

Proof. Write $x_i = \ell_i + (u_i - \ell_i)p_i$. Independently choose $X_i = u_i$ with probability p_i and $X_i = \ell_i$ otherwise. Multiaffinity gives $\mathbb{E}X = x$ and $\mathbb{E}f(X) = f(x)$. Every graph point therefore belongs to the convex hull of vertex graph points, and the reverse inclusion in the full graph hull is immediate. The vertical sections give the formulas. The feasible distributions form a nonempty compact polytope, so both extrema are attained. The lower and upper surfaces of the vertex graph polytope are respectively maxima and minima of finitely many affine functions on B , which also proves continuity at its boundary. $\square$

These representations are classical [3, Section 3]. They expose the compatibility issue: different factors may use different minimizing distributions in (2.2), whereas (2.4) requires one distribution for their sum.

For $m_e(x) = \prod_{i \in e} x_i$ with $e \neq \emptyset$, the unit-cube formulas are

$$\text{cav } m_e(x) = \min_{i \in e} x_i, \quad \text{vex } m_e(x) = \max\{0, \sum_{i \in e} x_i - |e| + 1\}. \quad (2.6)$$

To prove them, interpret $\mathbb{E}m_e(X)$ as the probability of joint success. The upper bound is attained by $X_i = \mathbf{1}\{U \leq x_i\}$ for one uniform $U \in [0, 1]$. For the lower bound, place failure intervals of lengths $1 - x_i$ consecutively on the unit circle. If their total length is below one they are disjoint, and if it is at least one they cover the circle. Their union therefore has measure $\min\{1, \sum_{i \in e} (1 - x_i)\}$. Each interval has its prescribed measure, so the complement gives the claimed minimum joint-success probability. Unused coordinates can be sampled independently to complete the mean vector.

Proposition 2.2 (Common upper attainment and deficiencies). *Let $f(x) = \sum_e a_e m_e(x)$ on $[0, 1]^n$, with $a_e \geq 0$. Discard supports of size at most one. For each remaining support choose an anchor i_e of smallest mean and put*

$$u_e = x_{i_e}, \quad S_e = \sum_{j \in e \setminus \{i_e\}} (1 - x_j), \quad t_e = \min\{u_e, S_e\}, \quad D_e(X) = X_{i_e} - \prod_{i \in e} X_i.$$

Then D_e is the nonnegative indicator that the anchor succeeds and some other support coordinate fails. Moreover,

$$\text{cav } f(x) = \sum_e a_e u_e, \quad (2.7)$$

$$T(f; x) = \sum_e a_e t_e, \quad (2.8)$$

$$H(f; x) = \max_{P \in \mathcal{P}_{[0,1]^n}(x)} \sum_e a_e \mathbb{E}_P D_e(X). \quad (2.9)$$

Proof. The common-threshold law attains every single-product upper envelope simultaneously. Positivity gives (2.7), and subtracting the two single-product envelopes gives (2.8). Finally, every admissible law has $\mathbb{E}X_{i_e} = u_e$, so minimizing the expected polynomial value is equivalent to maximizing the weighted sum of deficiencies. $\square$

The common-upper statement is classical [3, Theorems 4–5 in the authors’ manuscript]. A distribution used to prove a gap bound must work for all terms together; independently optimizing the distribution for each term merely recovers the termwise relaxation. Mixtures of admissible laws preserve the means, and nonnegative deficiencies allow the guarantees of different laws to be added.

Lemma 2.3 (Two classes handled by independence). *Put $c_0 = 1 - e^{-1}$. Independent Bernoulli rounding gives $\mathbb{E}D_e \geq c_0 t_e/2$ if either every support mean exceeds $1/2$, or at least two support means are at most $1/2$.*

Proof. Independence gives $\mathbb{E}D_e = u_e(1 - \prod_{j \neq i_e} x_j)$. In the second case a nonanchor mean is at most $1/2$, giving at least $u_e/2 \geq t_e/2$. In the first case the expression is at least $u_e(1 - e^{-S_e}) \geq c_0 u_e \min(1, S_e)$. If $S_e \leq 1$, this is at least $c_0 S_e/2 \geq c_0 t_e/2$; otherwise it is at least $c_0 u_e = c_0 t_e$. $\square$

Proposition 2.4 (Nonnegative-box transfer). *For a positive multilinear polynomial on B with $0 \leq \ell_i < u_i$, substitute $x_i = \ell_i + (u_i - \ell_i)p_i$ and expand the original monomials. The unit-cube polynomial f has nonnegative coefficients and no larger degree. With its expanded monomials as factors,*

$$H_B(f; x) = H(\tilde{f}; p), \quad T_B(f; x) \leq T(\tilde{f}; p). \quad (2.10)$$

One common-threshold endpoint law attains the upper envelope of every original monomial and their sum.

Proof. Expansion coefficients are products of nonnegative endpoints and interval widths. The full-envelope equality follows from the affine bijection. For an original factor expanded as $\sum_q g_q$, the sum of the lower convex envelopes is a convex underestimator and the sum of the upper concave envelopes is a concave overestimator. Subtract and sum over the original factors to obtain (2.10). Common-threshold rounding attains all expanded upper envelopes together and therefore also those of each original factor and their sum. $\square$

Expansion is a proof device and need not be an efficient representation. It changes the factor incidence structure. Degree bounds transfer through (2.10); bounds based on the original incidence graph require separate arguments. For bilinear functions, arbitrary box endpoints cause no such difficulty: an affine change introduces only scaled bilinear terms and affine summands, regardless of signs.

2.3 What a spatial certificate counts

Let $\mathcal{F}$ be a nonempty compact feasible set, let f be continuous, and put $f^* = \min_{x \in \mathcal{F}} f(x)$. A region oracle assigns a valid lower bound $\text{LB}(S)$ to feasible points in each region S . A *certificate at target* T_* is a finite cover $S_1, \dots, S_N$ of $\mathcal{F}$ with $\text{LB}(S_j) \geq T_*$ for all j . Together with a feasible value U and an absolute tolerance $\epsilon \geq 0$, the target $U - \epsilon$ certifies absolute gap at most ϵ . For $U > 0$ and a relative tolerance $0 \leq \theta < 1$, the target $(1 - \theta)U$ certifies $(U - f^*)/U \leq \theta$. A size lower bound may grant the best incumbent $U = f^*$, requiring $f^* > 0$ in this relative convention. Under these tolerance ranges, either target is nondecreasing in U , so worse incumbents require weakly higher targets.

A complete spatial split tree supplies a cover by its final regions. A lower bound for arbitrary covers is thus independent of branching order and node selection within the specified region and oracle model. If a procedure discards feasible regions by propagation or additional cuts, those deductions must also be included in the certificate model. A lower bound for one oracle does not automatically survive stronger coupled cuts, different branch coordinates, or analytic solution of separate components.

3 Signed bilinear gaps and interaction density

Let $G = (V, E)$ be a finite simple graph and $b_a(x) = \sum_{ij \in E} a_{ij} x_i x_j$ on $[0, 1]^V$, with arbitrary real coefficients. Zero-coefficient edges may be removed. Let $c^*(a)$ be the least $c \geq 0$ with $T(b_a; x) \leq cH(b_a; x)$ everywhere, and set $c^*(0) = 0$. Define

$$\rho_G(G) = \max_{\emptyset \neq W \subseteq V} \frac{|E(G[W])|}{|W|}, \quad (3.1)$$

with value zero for an empty graph. This is half the maximum subgraph average degree, not fractional arboricity, whose denominator is $|W| - 1$. Let d_G be degeneracy and Δ_G maximum degree. Then

$$d_G/2 \leq \rho_G(G) \leq d_G, \quad \rho_G(G) \leq \Delta_G/2. \quad (3.2)$$

The first lower bound uses a subgraph of minimum degree d_G ; the upper bounds use a degeneracy orientation and the degree sum identity.

Theorem 3.1 (Bilinear upper bounds). *For every real coefficient vector on G ,*

$$c^*(a) \leq \min\{4\sqrt{\rho_G(G)}, 4\sqrt{d_G}, 2\sqrt{\Delta_G}\}. \quad (3.3)$$

If G is bipartite with parts P, Q and has an edge, let Δ_P and Δ_Q be the maximum degrees in the respective parts. Then

$$c^*(a) \leq \sqrt{2 \min\{\Delta_P, \Delta_Q\}} \leq \sqrt{2 \min\{|P|, |Q|\}}. \quad (3.4)$$

The factor $\sqrt{2}$ in the first bipartite bound is sharp. The conclusions transfer to arbitrary finite boxes after removing fixed coordinates.

The proof refines the cut argument of Boland et al. [1]. The density order also follows from older operator theory, as shown in Section 3.6; no new general graph-norm principle is claimed.

3.1 An exact characterization by induced cuts

For $W \subseteq V$, set

$$Q_{a,W}(s) = \sum_{ij \in E(G[W])} a_{ij} s_i s_j, \quad L_W(a) = \sum_{ij \in E(G[W])} |a_{ij}|,$$

and define

$$R_W(a) = \frac{1}{2} \left(\max_{s \in \{-1,1\}^W} Q_{a,W}(s) - \min_{s \in \{-1,1\}^W} Q_{a,W}(s) \right). \quad (3.5)$$

A cut encoded by s has weight $\frac{1}{2} \sum_{ij} a_{ij} - \frac{1}{2} Q_{a,W}(s)$, so R_W is exactly the maximum cut weight minus the minimum cut weight. If $L_W > 0$ then $R_W > 0$: distinct degree-two sign characters are orthogonal under uniform signs, and their nonzero combination cannot be constant. Set $L_W/R_W = 0$ when both vanish.

Lemma 3.2 (Induced-cut characterization). *For every coefficient vector,*

$$c^*(a) = \max_{W \subseteq V} \frac{L_W(a)}{R_W(a)}. \quad (3.6)$$

Each nonzero induced ratio is attained at $x_i = 1/2$ on W and $x_i = 0$ outside W .

Proof. At that point any admissible vertex law fixes outside coordinates to zero. Write $X_i = (1+s_i)/2$ on W . Since $\mathbb{E}s_i = 0$,

$$\mathbb{E}b_a(X) = \frac{1}{4} \left(\sum_{ij \in E(G[W])} a_{ij} + \mathbb{E}Q_{a,W}(s) \right).$$

An extremizing sign vector and its negative, each of probability $1/2$, attain either extreme of $Q_{a,W}$ with the prescribed means. Thus $H = R_W/2$, whereas each active product has width $1/2$ and

$T = L_W/2$. These identities hold at every point of $\{0, 1/2, 1\}^V$, with W its set of half-valued coordinates: fixing coordinates to one instead of zero adds only affine terms on W .

To extend the comparison, the width of the product $x_i x_j$ is

$$\min(x_i, x_j) - \max(0, x_i + x_j - 1) = \min(x_i, x_j, 1 - x_i, 1 - x_j).$$

Partition the cube by the hyperplanes $x_i = x_j$, $x_i + x_j = 1$ ($i \neq j$), and $x_i = 1/2$. The termwise gap is affine on every closed cell. Every cell vertex is in $\{0, 1/2, 1\}^V$. Indeed, active equations fix a coordinate to 0, 1/2, or 1, or relate two coordinates by equality or complementation. In a connected component of these relations, one fixed coordinate determines all values. If none is fixed, a cycle with an odd number of complementations forces the value 1/2. Otherwise a free parameter remains and a small perturbation preserves all active equations, contradicting extremality. This accounts for every cell vertex.

The graph-hull gap is concave, being $\text{cav } b_a - \text{vex } b_a$. Hence $T - cH$ is convex on each cell for $c \geq 0$. Its value at any convex combination of cell vertices is at most the combination of its values there. The grid-point bound therefore proves the upper direction of (3.6), and the attained ratios prove the other. $\square$

The grid identities and upper implication are those of Boland et al. [1, Lemma 1 and Corollary 1], following Luedtke et al. [3]. The cell argument above supplies the boundary details.

3.2 Cut ranges, row norms, and fractional orientations

Let A be the symmetric weighted adjacency matrix on W , with zero diagonal and zero entries at nonedges. Write $a_{i,T} = (A_{ij})_{j \in T}$ for $T \subseteq W$ and $a_i = a_{i,W}$. For a rectangular matrix define $\|M\|_{\infty \rightarrow 1} = \max_u \text{sign} \|Mu\|_1 = \max_{u,v} \text{sign } v^\top Mu$; empty matrices have norm zero.

Lemma 3.3 (Polarization and row estimates). *Writing $R = R_W(a)$,*

$$R = \max_{S \subseteq W} \|A_{S,W \setminus S}\|_{\infty \rightarrow 1}, \quad (3.7)$$

$$R \geq \frac{1}{\sqrt{2}} \max_{S \subseteq W} \sum_{i \in S} \|a_{i,W \setminus S}\|_2, \quad (3.8)$$

$$R \geq \frac{1}{4} \sum_{i \in W} \|a_i\|_2. \quad (3.9)$$

Proof. For signs s, t , put $u = (s + t)/2$, $v = (s - t)/2$. Their supports are complementary, and symmetry gives $Q(s) - Q(t) = 2v^\top Au$. Conversely, arbitrary signs on complementary supports arise from $s = u + v$, $t = u - v$. Maximizing half the difference proves (3.7).

The sharp real Khinchin inequality of Szarek [5] states that independent uniform signs satisfy

$$\mathbb{E} \left| \sum_j c_j \varepsilon_j \right| \geq \frac{1}{\sqrt{2}} \left(\sum_j c_j^2 \right)^{1/2}. \quad (3.10)$$

Apply it to every row of $A_{S,T}$ for a fixed partition S, T , and sum. The best sign vector performs at least as well as its average, proving (3.8). Transposition gives the same estimate for side T .

Choose a partition locally maximizing the cut weight with nonnegative weights a_{ij}^2 . At every vertex at least half its squared incident weight crosses, since otherwise moving the vertex improves the cut. Thus

$$X = \sum_{i \in S} \|a_{i,T}\|_2, \quad Y = \sum_{i \in T} \|a_{i,S}\|_2 \quad \text{satisfy} \quad X + Y \geq \frac{1}{\sqrt{2}} \sum_i \|a_i\|_2.$$

The two row estimates yield $R \geq \max(X, Y)/\sqrt{2} \geq (X + Y)/(2\sqrt{2})$, which proves (3.9). $\square$

Lemma 3.4 (Fractional orientation). *Every finite graph H admits nonnegative fractions $\theta_{ij} + \theta_{ji} = 1$ on each edge such that $\sum_j \theta_{ij} \leq \rho_G(H)$ for every vertex. This is the least possible uniform bound on the vertex loads.*

Proof. For $m = |E(H)| > 0$, form a flow network with a source, edge nodes, vertex nodes, and a sink. Source-to-edge capacities are one; arcs from an edge node to its endpoints have capacity $m + 1$; vertex-to-sink capacities are t . A cut of capacity at most m cannot cut an edge-to-endpoint arc. For a fixed set U of vertex nodes on the source side, placing all edge nodes internal to U on that side minimizes capacity, giving $m - |E(H[U])| + t|U|$. At $t = \rho_G(H)$ every cut has capacity at least m . Max-flow/min-cut supplies a flow saturating the source arcs; outgoing flows at each edge define its fractions. Conversely, the loads over U sum to at least $|E(H[U])|$, proving minimality. For $m = 0$ use zero loads. $\square$

Proof of Theorem 3.1. Fix an induced graph H with nonempty weighted support and abbreviate $L = \sum_{ij} |a_{ij}|$, $R = R_{V(H)}(a)$. Weighted Cauchy–Schwarz gives

$$\sum_j \theta_{ij} |a_{ij}| \leq \left(\sum_j \theta_{ij} \right)^{1/2} \left(\sum_j \theta_{ij} a_{ij}^2 \right)^{1/2} \leq \sqrt{\rho_G(H)} \|a_i\|_2.$$

Summing counts each absolute edge weight once. By (3.9), $L \leq 4\sqrt{\rho_G(H)}R$. Also $\|a_i\|_2 \geq \|a_i\|_1/\sqrt{\Delta_H}$ at every nonisolated vertex, and $\sum_i \|a_i\|_1 = 2L$. The same row bound therefore gives $R \geq L/(2\sqrt{\Delta_H})$. The graph parameters do not increase in induced subgraphs, so Lemma 3.2 proves (3.3).

For bipartite H use its inherited parts P', Q' in (3.8). Every edge crosses, giving

$$R \geq \frac{1}{\sqrt{2}} \sum_{i \in P'} \|a_i\|_2 \geq \frac{L}{\sqrt{2\Delta_P}}.$$

Transpose to obtain the bound with Δ_Q and apply the same induced characterization. On a unit-weight four-cycle with one negative edge, $L = 4$, $R = 2$, and $\Delta_P = \Delta_Q = 2$. Its center ratio attains two, so the stated bipartite constant is sharp. An affine box transformation scales each bilinear coefficient by the two interval widths and introduces only affine summands, proving the final assertion. $\square$

The degeneracy bound has a distinct direct proof. Orient along a degeneracy order with outdegree at most d_G and color vertices in reverse order. Place a vertex opposite the side receiving at least half the squared weight of its outgoing edges. Its crossing row norm is at least its outgoing row norm divided by $\sqrt{2}$. Summing and applying Cauchy–Schwarz to at most d_G outgoing entries gives $X + Y \geq L/\sqrt{2d_G}$. The final step of Lemma 3.3 yields $R \geq L/(4\sqrt{d_G})$. This proof is weaker than the density theorem but uses an ordinary acyclic orientation.

3.3 Sharp order on every interaction graph

Define

$$\Gamma(G) = \sup_{a \in \mathbb{R}^E \setminus \{0\}} c^*(a), \quad \Gamma_{\pm}(G) = \max_{a \in \{-1, 1\}^E} c^*(a),$$

with both values zero for edgeless graphs.

Theorem 3.5 (Worst coefficients on a fixed graph). *For every graph with an edge,*

$$\max \left\{ 1, \sqrt{\frac{\rho_G(G)}{2 \log 2}} \right\} \leq \Gamma_{\pm}(G) \leq \Gamma(G) \leq 4\sqrt{\rho_G(G)}. \quad (3.11)$$

Requiring every coefficient to be nonzero does not change $\Gamma(G)$. It also equals

$$\sup_{a \neq 0} L_V(a)/R_V(a). \quad (3.12)$$

The lower bound has a full-support signing and a witness in $\{0, 1/2\}^V$.

Proof. On an h -vertex, m -edge graph with $m > 0$, choose independent edge signs σ_e . For fixed vertex signs s and $\lambda > 0$, $\mathbb{E}e^{\lambda Q_{\sigma}(s)} = \cosh(\lambda)^m \leq e^{m\lambda^2/2}$, and similarly for $-\lambda$. Put $Z = \max_s |Q_{\sigma}(s)|$. Global sign reversal preserves Q_{σ} , so

$$e^{\lambda Z} \leq \sum_{s: s_1=1} (e^{\lambda Q_{\sigma}(s)} + e^{-\lambda Q_{\sigma}(s)}).$$

Taking expectations and using Jensen gives $\mathbb{E}Z \leq h \log 2/\lambda + m\lambda/2$. At $\lambda = \sqrt{2h \log 2/m}$, at least one signing satisfies

$$\|Q_{\sigma}\|_{\infty} \leq \sqrt{2mh \log 2}. \quad (3.13)$$

The values at different sign vectors need not be independent.

Apply this to a densest induced graph $H = G[W]$. Since $L_W = m$ and $R_W \leq \|Q_{\sigma}\|_{\infty}$, $L_W/R_W \geq \sqrt{\rho_G(G)/(2 \log 2)}$. Give all remaining edges arbitrary signs; they do not affect the witness face of Lemma 3.2. The lower bound one follows from $R_W \leq L_W$, and the upper bound is Theorem 3.1.

Call the supremum in (3.12) C . The full center shows $\Gamma \geq C$. Conversely, every induced ratio is a whole-graph ratio after zeroing all coefficients outside the induced subgraph. Thus $\Gamma \leq C$. On the compact set $L_V(a) = 1$, R_V is continuous and strictly positive, so the ratio attains a maximum. Perturb zero entries of a maximizer to nonzero values. Continuity of the full-graph ratio shows that the full-support supremum is also C . This does not assume continuity of c^* as edges disappear or attainment under the full-support restriction. $\square$

Uniform boundedness of all signed factors in a graph family is therefore equivalent to uniform boundedness of its induced density. No closure assumption on the family is needed: fixing outside coordinates exposes a densest subgraph. Global average density is insufficient. Attaching q^2 leaves to one vertex of $K_{q,q}$ produces a connected graph with total edge-to-vertex ratio less than two but induced density at least $q/2$.

The density is unweighted. Scaling all coefficients by a nonzero scalar preserves the gap ratio, whereas a sum of coefficient magnitudes scales. A single edge of arbitrarily small magnitude therefore rules out replacing ρ_G by $\max_W \sum_{e \in E(W)} |a_e|/|W|$. The row estimates remain coefficient-sensitive certificates; a uniform gap bound must use them on every induced subgraph, not only the full graph.

The order may equivalently be expressed using degeneracy or arboricity. If α_G is the minimum number of forests partitioning the edges, then $\rho_G(G) \leq \alpha_G \leq d_G \leq 2\rho_G(G)$. For the middle inequality, label outgoing edges in a degeneracy orientation with distinct labels from 1 to d_G . Each label class is a forest: an undirected cycle would have an earliest vertex with two outgoing edges of that label. The other inequalities follow by counting forest edges and (3.2).

3.4 Positive coefficients, exactness, and finite constants

For positive coefficients a proper coloring with $\chi \geq 2$ colors gives the classical stronger bound

$$c^*(a) \leq \begin{cases} 2 - 2/\chi, & \chi \text{ even,} \\ 2 - 2/(\chi + 1), & \chi \text{ odd.} \end{cases} \quad (3.14)$$

Choose uniformly $\lfloor \chi/2 \rfloor$ colors for one side of a cut. Every edge crosses with probability $p_\chi = 2\lfloor \chi/2 \rfloor \lceil \chi/2 \rceil / [\chi(\chi - 1)]$. Thus the maximum cut is at least $p_\chi L$, and the minimum cut is zero. The same coloring works on every induced subgraph, so Lemma 3.2 gives (3.14). This is the bound of Luedtke et al. [3, Theorem 8 in the authors' manuscript]. In particular, the positive bilinear factor is at most two and is one when its nonzero support is nonempty and bipartite. For the all-positive K_n with $n \geq 2$, the center ratio is $\binom{n}{2} / \lfloor n^2/4 \rfloor$, tending to two. Hence the worst-over-signs quantifier in Theorem 3.5 is essential.

Assume the effective support graph, obtained by deleting zero-coefficient edges, has at least one edge. Then $c^*(a) = 1$ if and only if every cycle of this graph has an even number of positive and an even number of negative edges [1, Theorem 4]. Boland et al. identify this as a consequence of the convex-envelope criterion in Theorem 3.10 of Misener, Smadbeck, and Floudas, *Dynamically generated cutting planes for mixed-integer quadratically constrained quadratic programs and their incorporation into GloMIQO 2*; our attribution to that predecessor is through Boland et al., who give the alternative cut proof used here. Indeed, $R_V = L_V$ exactly when one cut contains all positive edges and no negative edges, and another contains all negative edges and no positive edges. An edge set is a cut exactly when it intersects every cycle evenly. Necessity follows by traversing a cycle. For sufficiency, label vertices by the parity of specified edges on a path from a root in each component; cycle parity makes the label path-independent and supplies the cut. The criterion passes to induced subgraphs, proving sufficiency for $c^* = 1$. Conversely, $c^* = 1$ implies $R_V = L_V$ by Lemma 3.2. In particular, every forest with at least one edge has factor one for every signing. For empty effective support the gaps coincide at zero and $c^* = 0$ by convention.

The best universal constant in $\Gamma(G) \leq K_\rho \sqrt{\rho_G(G)}$ lies in $[2, 4]$: the lower bound is the four-cycle with one negative edge, with density one and factor two. For degeneracy the corresponding interval is $[\sqrt{2}, 4]$. These are finite-parameter questions, distinct from the best constants along sequences with density or degeneracy tending to infinity; the four-cycle does not give a growing-density lower bound.

The bounds imply $c^* \leq 4\sqrt{2}$ for series-parallel graphs by degeneracy, $c^* \leq 4\sqrt{3}$ for planar graphs by density, and $c^* \leq \sqrt{2 \min(p, q)}$ for $K_{p, q}$. For K_n , the maximum-degree bound is $2\sqrt{n - 1}$, compared with $600\sqrt{n}$ in Boland et al. [1, Theorem 2]. These are scalar vertical-gap comparisons, not approximation guarantees for a constrained objective.

3.5 Exact finite full-signing constants

For $n \geq 2$, distinguish the largest full-center ratio among complete-graph signings from the largest ratio over all points:

$$M_n = \max_{a \in \{-1, 1\}^{E(K_n)}} \frac{\binom{n}{2}}{R_V(a)}, \quad F_n = \Gamma_\pm(K_n).$$

All edges here have magnitude one. These maxima differ from the arbitrary-real-coefficient supremum in (3.12).

Proposition 3.6 (Finite exhaustive computation). *The exact values are*

n	3	4	5	6	7
$\min_a R_V(a)$	2	4	4	5	8
M_n	3/2	3/2	5/2	3	21/8
F_n	3/2	3/2	5/2	3	3

Proof. The first row is established by the exhaustive integer computation in Appendix A, which proves completeness of the enumeration, gives executable code, and supplies attaining signings. Dividing $\binom{n}{2}$ by that row gives M_n . Lemma 3.2 gives

$$F_n = \max_{2 \leq k \leq n} M_k.$$

Indeed, each induced support of a full signing is a full signing of some K_k , giving the upper bound. Conversely, any signing of K_k extends to a full signing of K_n by assigning signs to all remaining edges, and its ratio persists on the face fixing the new coordinates to zero. Since $M_2 = 1$, this proves the last row. $\square$

In particular $M_7 < M_6$ while $F_7 = F_6 = 3$. An explicit witness is the K_6 signing in Table 1, extended by a new vertex with all incident coefficients positive. At the point whose six old coordinates are $1/2$ and whose new coordinate is zero, its ratio is three; at the full center its ratio is only $21/10$. The finite computation optimizes precisely these listed full-signing problems; it makes no claim about larger n or the optimal constants for arbitrary real coefficients.

3.6 The relation to Schur-multiplier theory

We record the older implication to separate the density order from the explicit elementary constant. For a real matrix M , its Schur multiplier acts by entrywise multiplication, with norm $\|M\|_{\text{Schur}}$ measured in the Euclidean operator norm. Define the real projective norm

$$\pi(M) = \inf \left\{ \sum_k \|u_k\|_\infty \|v_k\|_\infty : M = \sum_k u_k v_k^T \right\}.$$

The established results of Davidson and Donsig [2, Theorems 1.2 and 2.4] give

$$\|M\|_{\text{Schur}} \leq 2\sqrt{\beta(P)}, \quad \pi(M) \leq K_G \|M\|_{\text{Schur}}, \quad (3.15)$$

for $|M_{ij}| \leq 1$ supported on a pattern P , where K_G is the real Grothendieck constant and

$$\beta(P) = \max_{R, C: |R|+|C|>0} \frac{|P \cap (R \times C)|}{|R| + |C|}.$$

The first bound uses the weighted matrix theorem, which permits continuous row and column bounds. Using the integer pattern theorem instead would introduce rounding. These are established norm inequalities, not statements about sizes of convex extended formulations.

For the symmetric adjacency pattern,

$$|P \cap (R \times C)| \leq |E(G[R \cup C])| + |E(G[R \cap C])| \leq \rho_G(G)(|R| + |C|).$$

An edge is counted at most once unless both endpoints lie in $R \cap C$, when both matrix orientations are counted. Taking $R = C$ equal to a densest set proves $\beta(P) = \rho_G(G)$. For weighted adjacency A , take $M = \text{sign}(A)$, with zeros where A vanishes. The definition of π gives

$$2L_V = \langle A, M \rangle \leq \pi(M) \|A\|_{\infty \rightarrow 1}.$$

For sign vectors u, v , put $p = (u+v)/2$, $q = (u-v)/2$. Symmetry yields $u^\top Av = 2[Q_{a,V}(p) - Q_{a,V}(q)]$. Multiaffinity puts these two values between the vertex extrema on $[-1, 1]^V$, so $\|A\|_{\infty \rightarrow 1} \leq 4R_V$. Consequently

$$L_V \leq 4K_G \sqrt{\rho_G(G)} R_V. \quad (3.16)$$

Apply this on every induced graph to recover the density upper order. The direct random-sign argument supplies the matching bilinear lower order.

Equivalently, let $S(G) = \sup_{a \neq 0} L_V(a) / \|Q_{a,V}\|_\infty$ be the real Sidon constant of the edge-supported degree-two sign characters, with $S(G) = 0$ for edgeless graphs. Because Q has mean zero, $\|Q\|_\infty / 2 \leq R_V \leq \|Q\|_\infty$ and hence $S(G) \leq \Gamma(G) \leq 2S(G)$. For bipartite graphs, flipping all signs in one part negates Q , making its range symmetric and giving $S(G) = \Gamma(G)$. The constant four improves the particular transfer (3.16); it is not asserted to be the best constant available in all prior literature.

A Exhaustive verification of the finite signing constants

This appendix gives the finite computational proof used in Proposition 3.6. Label vertices $0, \dots, n-1$. For vertex signs t_i , switching replaces a_{ij} by $a'_{ij} = a_{ij}t_it_j$. The identity $Q_{a'}(s) = Q_a(t \cdot s)$ shows that switching preserves both extrema of Q , and hence R_V . Taking $t_0 = 1$ and $t_j = a_{0j}$ makes every coefficient incident to zero positive. This normalized representative is unique within each switching class: a switching that keeps those coefficients positive has $t_j = t_0$ for all j , so changes no edge. Thus it suffices to enumerate the $2^{\binom{n-1}{2}}$ choices of negative edges among vertices $1, \dots, n-1$. There are respectively 2, 8, 64, 1024, 32768 representatives for $n = 3, \dots, 7$.

Encode a cut by the subset on side one, always leaving vertex zero on side zero. Complementation changes no cut, so the 2^{n-1} such subsets include every cut. If a cut has c crossing edges, of which h are negative, its weight is exactly $c - 2h$. The following complete Python 3.10+ program computes the first row of Proposition 3.6 using unbounded integers. The operation `bit_count()` counts set bits in an integer; each free-edge mask records which edges cross the cut. No floating-point arithmetic or solver is used.

```
from itertools import combinations
def min_range(n):
    edges = list(combinations(range(n), 2))
    free = [(i, j) for i, j in edges if i != 0]
    cuts = []
    for side in range(1 << (n - 1)):
        side <= 1 # Vertex zero stays on side zero.
        cross = lambda i, j: ((side >> i) ^ (side >> j)) & 1
        size = sum(cross(i, j) for i, j in edges)
        mask = sum(cross(i, j) << k
                    for k, (i, j) in enumerate(free))
        cuts.append((size, mask))
    best = len(edges) + 1
    for negative in range(1 << len(free)):
        weights = [size - 2 * (negative & mask).bit_count()
                    for size, mask in cuts]
        best = min(best, max(weights) - min(weights))
    return best
result = [min_range(n) for n in range(2, 8)]
assert result == [1, 2, 4, 4, 5, 8]
print(result)
```

Every signed cut weight lies between $-\binom{n}{2}$ and $\binom{n}{2}$; its range is at most $\binom{n}{2}$, because two cut weights differ by a sum of at most that many coefficients in $\{-1, 1\}$. The initial value `len(edges) + 1` therefore exceeds every candidate range. The two loops cover all normalized signings and all cuts, so their minimum is exactly $\min_a R_V(a)$, rather than merely an upper bound supplied by the best witness found. This is an exhaustive finite proof, not an analytic formula for general n .

n	Negative edges; all other edges have coefficient +1	$\min Q$	$\max Q$
3	$\emptyset$	-1	3
4	$\emptyset$	-2	6
5	12, 14, 23	-4	4
6	14, 15, 23, 25, 34	-5	5
7	12, 13, 16, 23, 25, 34	-7	9

Table 1: Attaining full-center signings, with vertices numbered from zero. An entry ij denotes the unordered pair $\{i, j\}$.

The extrema in Table 1 can be checked by direct evaluation of Q on the 2^{n-1} vertex-sign vectors with $s_0 = 1$. They give $R_V = (\max Q - \min Q)/2$ equal to the computed minimum in each case. Extending the listed K_6 witness by vertex six and positive new edges gives $\min Q = -9$, $\max Q = 11$ at the full center, hence ratio 21/10; its original six-vertex face retains ratio three.

The accompanying script `verification/check_complete_signings.py` implements the same integer enumeration, records a histogram of all candidate ranges, and checks the displayed witnesses by direct evaluation of Q on every induced face. Running it with Python 3.10+ writes `verification/complete_signings.json`, including the witnesses, counts, exact rational ratios, and the extended six-vertex example.

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